



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI SUMANGALI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

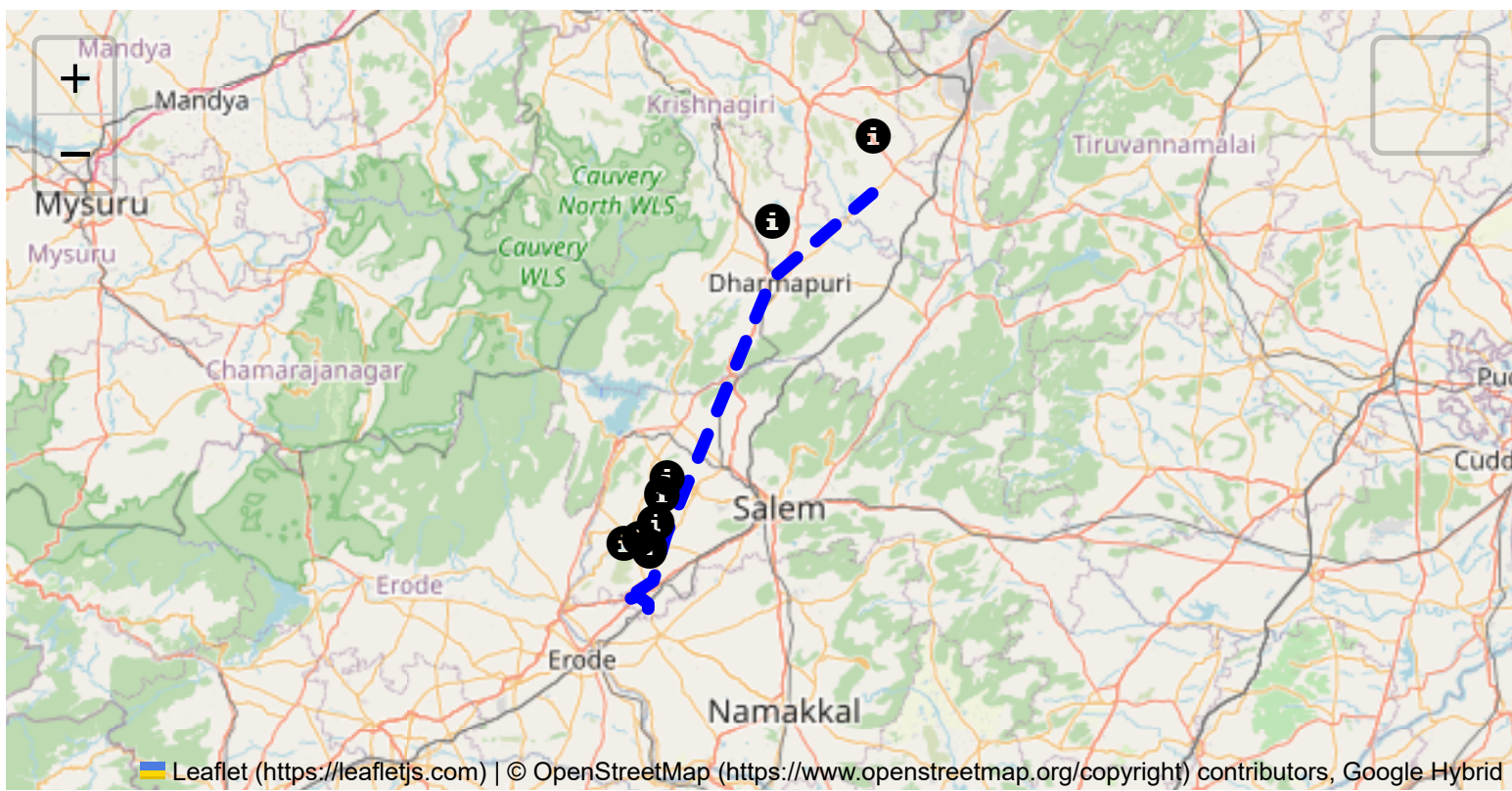
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 138.51 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.325979, 78.360019)



Welcome to the Journey Risk Management Study

Here's an analysis of the route from CVQF+23W, Sangagiri to 89G5+FVF, Kullalur in Tamil Nadu, focusing on potential hazards and considerations for a truck driver carrying hazardous materials:

1. Overview of the Route Map:

- The route extends approximately 138.51 kilometers and passes through various towns and smaller villages in Tamil Nadu, primarily via state highways such as SH 86.

- Key waypoints include Sankari RS, Sankari West, Padaiveedu, Devannagoundanur, Ayveli, Erumaipatti, Konganapuram, Koranampatti, and Pennagaram.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate, with hot summers (March to June) and monsoon rains (June to September and November to December).
- Potential weather-related hazards include heavy rainfall during monsoon seasons, possibly causing poor visibility and slippery roads, especially toward Pennagaram, which is closer to hilly areas.

3. Analysis of Traffic Patterns:

- Morning and evening peak hours might show increased traffic due to local commuting, especially near populated areas like Sankari.
- Congestion-prone areas include towns and market roads where local activities might slow down traffic.

4. Assessment of Road Quality and Infrastructure:

- Roads vary in quality, with state highways generally maintained but rural roads potentially having potholes or being narrow.
- Bridges and underpasses should be checked for height and weight restrictions, although major roads are typically suitable for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies:

- In case of blockages, alternative routes could involve detours via NH 544 for a faster, although longer detour toward Erode and Krishnagiri roads.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Compliance with Tamil Nadu Motor Vehicles Rules and national regulations for hazardous materials transport is mandatory.
- Permits and route approvals might be necessary, especially for materials with specific transport restrictions.

7. Overview of Historical Incidents:

- Although specific historical incident data isn't available, routes like SH 86 often experience road accidents due to sharp turns and poor visibility. Local news archives could provide more detailed reports.

8. Environmental Considerations and Sensitive Areas:

- Proximity to water bodies and agricultural lands should be noted, with extra caution to prevent contamination in case of a spill.
- Areas near Pennagaram may be closer to forested or wildlife-protected areas.

9. Analysis of Communication Coverage:

- Major network coverage should be available throughout the route, but possible dead zones in more remote sections like Erumaipatti to Konganapuram.
- It's advised to carry communication backups such as satellite phones if possible.

10. Estimated Emergency Response Times:

- Urban-adjacent areas like Sangagiri and Sankari typically have faster emergency access due to proximity to larger towns.
- Rural segments could experience delays, with response times increasing particularly toward less populated areas like Kullanur.

11. Overall Summary of Risk Assessment:

- The route is mostly suitable for transportation of hazardous materials if precautions are taken regarding weather, peak traffic, and road conditions.
- Key risks include potential road surface conditions in rural areas, weather impacts, and heightened alertness in environmentally sensitive zones.
- Preparedness in terms of communication and OPR (Operational Preparedness and Response) plans greatly lowers risk impact.

For a successful and risk-mitigated journey, adherence to regulatory compliance and constant communication with logistics operations is essential.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45764, 77.85219	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45571, 77.81702	15 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.46916, 77.82431	30 KM/Hr
11	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
12	Turn	High	11.49465, 77.88488	15 KM/Hr
13	Turn	Medium	11.56096, 77.89931	30 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.59214, 77.90656	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	Medium	11.67870, 77.96846	30 KM/Hr
21	Turn	High	11.70227, 77.97864	15 KM/Hr
22	Turn	Medium	11.71715, 77.99969	30 KM/Hr
23	Turn	Medium	11.71921, 78.01133	30 KM/Hr
24	Turn	Medium	11.72737, 78.02248	30 KM/Hr
25	Turn	Medium	11.73775, 78.03465	30 KM/Hr
26	Turn	High	11.73651, 78.03878	15 KM/Hr
27	Turn	Medium	11.74189, 78.04660	30 KM/Hr
28	Turn	Medium	11.74309, 78.04725	30 KM/Hr
29	Turn	Medium	11.77961, 78.05775	30 KM/Hr
30	Turn	Medium	11.86873, 78.08374	30 KM/Hr
31	Turn	Medium	11.87966, 78.08187	30 KM/Hr
32	Turn	Medium	11.92609, 78.06513	30 KM/Hr
33	Turn	Medium	11.93719, 78.05372	30 KM/Hr
34	Turn	High	11.95828, 78.04983	15 KM/Hr
35	Turn	Medium	11.95997, 78.04946	30 KM/Hr
36	Turn	Medium	11.96335, 78.05141	30 KM/Hr
37	Turn	High	11.96408, 78.05131	15 KM/Hr
38	Turn	Medium	11.96511, 78.04966	30 KM/Hr
39	Turn	High	11.96608, 78.04919	15 KM/Hr
40	Turn	Medium	11.96819, 78.05088	30 KM/Hr
41	Turn	High	12.14053, 78.13964	15 KM/Hr
42	Turn	High	12.14497, 78.16507	15 KM/Hr
43	Turn	Medium	12.14707, 78.17538	30 KM/Hr
44	Turn	Medium	12.20764, 78.24739	30 KM/Hr
45	Turn	Medium	12.26801, 78.30453	30 KM/Hr
46	Turn	Medium	12.26983, 78.30724	30 KM/Hr
47	Turn	Medium	12.28372, 78.31950	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
31	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
32	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
33	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
34	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
35	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
36	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
38	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
39	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
40	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
41	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
42	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
43	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13658, 78.16171



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14589, 78.17156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86873, 78.08374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95828, 78.04983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96608, 78.04919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14053, 78.13964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14497, 78.16507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14707, 78.17538



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.20764, 78.24739



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26801, 78.30453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26983, 78.30724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.28372, 78.31950
